



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

We had better keep our programs safe.



Learning Outcome:

Understand important information about client software for the identity metasystem in spoken or written documents.

Ask your partner:

- Is your computer safe from any virus? Why?
- What measures have to be taken to keep our programs secure?

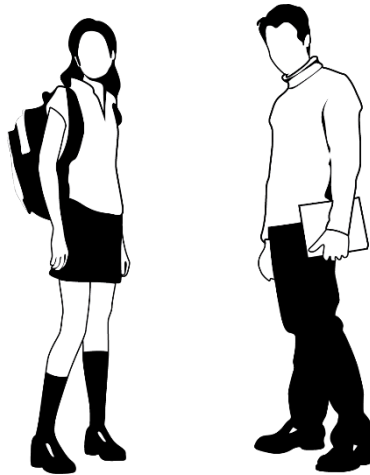
Can Do Statements:

- I can talk about the role of code access security infrastructure in the .NET framework.
- I can present features of code access security in the .NET framework.
- I can discuss information about developing an application in the .NET framework.



Let's Explore!

Computers are now more secure!



Instructions: Read the text below.

Sam: Hey, Lucy! How are you?

Sam: Hey, Lucy! How are you?

Lucy: Hi, Sam! I'm fine, but I'm really worried about the final test.

Sam: I know. I feel the same. We'd better study together so that we feel more confident.

Lucy: You're right, Sam. You know what? I have serious problems with security in the .NET Framework.

Sam: Well, It's not difficult. It simply improves the security in your computer with something called Code Access Security.

Lucy: That's exactly what's unclear to me. What is it?

Sam: It's an infrastructure that allows people to differentiate permissions or rights that a code has from the rights that you have. Remember that, in the past, every code on a computer had the same rights to access resources as you did as a user of the computer. It means there were no restrictions to access the information of the computer.

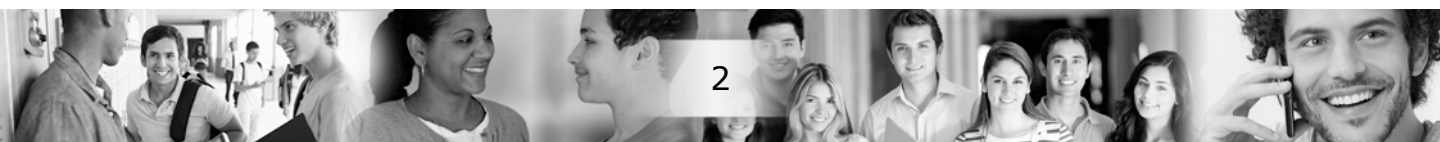
Lucy: I see. What are the characteristics of Code Access security?

Sam: It protects computer systems from malicious code and protects trusted code from accidentally compromising security; additionally, it gives you more control over the applications you want to perform by limiting the action of those on your computer. Program users might want to work in a virus-controlled environment.

Lucy: So, this affects the development of applications, right?

Sam: Sure. The first thing you ought to guarantee when developing an application is security. Then, you have to decide how the application will perform in a semi-trusted or less than a full-trust environment. These environments will need a different application design. The safest applications are created considering the permissions you grant as a programmer.

Lucy: Thanks, Sam! Now everything is clear to me. I hope I do well in the test.



**Let's Practice!**

Code Access Security and Applications

Instructions: Select the correct answer.

1. Is Code Access Security one of the best ways to improve the security in your computer?
 - a. No, it isn't.
 - b. No, it can't.
 - c. Yes, it is.
 - d. Yes, it can.
2. How can you differentiate the permissions or rights a code has?
 - a. With infrastructure
 - b. With Code Access Security
 - c. With an application
 - d. With a computer
3. Does a code have access to resources?
 - a. Yes, it has access to resources.
 - b. Yes, it had resources.
 - c. Yes, it was resources.
 - d. Yes, it have access to other networks.

**Let's Practice!**

We ought to use Code Access Security in our computer systems.

Instructions: Circle the correct words to complete the sentences.

1. Managers (had / has) better use Code Access Security.
2. People (are / ought) to protect their computer systems form malicious code.
3. Engineers (might / could) want to protect trusted code from accidentally compromising security.
4. You (ought / would) to guarantee security when developing an application.
5. Applications (ought / had) better perform in a semi-trusted environment.



Extension Activity

The importance of Code Access Security

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Write why Code Access Security is important.
- Check with your teacher the information that you wrote.
- Share the information with your classmates.

Self-Assessment

Yes

No

- I write about Code Access Security.
- I share the information about Code Access Security with my teacher.
- I talk about Code Access Security with my classmates.

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Target Words

- **trust** (noun): reliance on the security of something
e.g. Then, you have to decide how the application will perform in a semi-trusted or less than a full-trust environment.
- **malicious code** (noun): a kind of harmful computer code that makes the system vulnerable to damage or data theft
e.g. It protects computer systems from malicious code and protects trusted code from accidentally compromising security.
- **code** (noun): symbolic representation of statements or instructions in a computer program
e.g. It's an infrastructure that allows people to differentiate permissions or rights that a code has from the rights that you have.

- **compromise** (verb): to expose in danger
e.g. It protects computer systems from malicious code and protects trusted code from accidentally compromising security.
- **application** (noun): it's a program intended to perform a specific function directly for a user
e.g. The first thing you ought to guarantee when developing an application is security.
- **grant** (verb): to give or confer
e.g. The safest applications are created considering the permissions you grant as a programmer.



Learning Points

Had Better, Ought to, and Might Want To to Give Advice

Had better, ought to, and might want to are three of the different options you have to offer another person a piece of advice.

When you use those forms, it is not necessary to keep the subject-verb agreement for third person singular. See these examples:

Michael **had better use** the .NET framework.

Jessica **ought to** write a letter to ask for a change in the program.

You **might want to** ask for help if you have doubts using the program.

The structure of these forms are always the same regardless the subject.

Examples

- *We'd better study together so that we feel more confident.*
- *The first thing you ought to guarantee when developing an application is security.*

Degrees of Trust Granted on an Application

There is a difference between the code that runs on your computer and the code that is created to run on other machines.

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The degree of trust granted on an application depends on where the code resides and how it is started. When creating an application, it is important to be aware of the degrees of trust that are granted by default in each computer:

- 1. Full trust permission set:** This is the code that the local computer grants.
- 2. Local intranet permission set:** It is granted by the local network.
- 3. Internet permission set:** This code is granted by the internet.

These default values may affect the permissions to your application. This means that the application can work differently on each computer depending on the security policy; for instance, if your application requires an internet connection to update, the program will not update unless you have access to the internet. Otherwise, the user has to change the security settings to allow it.





Learning Points

Examples

- We'd better study together so that we feel more confident.
- The first thing you ought to guarantee when developing an application is security.

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